

Stellar subdivision schemes that split 2-faces, and typed-cycle hybrids with Maubach bisection

Research note (not part of the article)

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Abstract

We record a stellar subdivision scheme for 4-dimensional ordered meshes that splits a 2-face (a triangle) of each cell instead of an edge (Maubach) or a facet (Mitchell). With the right rule for the types of the children, the scheme is combinatorially correct: its invariants are Maubach's with the split edge $\{0, l\}$ replaced by the split face $\{0, 1, l\}$, and the proof of the article goes through verbatim. The scheme belongs to a family of *typed cycles*, for which we give three local conditions that are sufficient, and in exhaustive tests also necessary, for the invariants to be preserved. Geometrically, however, a scheme that splits only faces of dimension ≥ 2 never removes an edge, so cell diameters never tend to zero. Hybrid cycles mixing edge and face splits do shrink cells, but none of those we tested keeps shape regularity. All claims marked *verified* were checked by the scripts in `scripts/`.

1 Setting

We use the notation and framework of the article (*Stellar subdivision schemes and pointerless Maubach bisection on balanced seeds*, §3). A cell is an ordered d -simplex $\sigma = \langle \sigma[0], \dots, \sigma[d] \rangle$, ∂_i deletes the i -th vertex, and equalities $\partial_i \sigma = \partial_j \omega$ are equalities of *ordered* simplices. An ordered stellar subdivision $(\delta, v)_l$ of a face $\delta = \sigma[p_0, \dots, p_k]$ replaces every cell $\sigma \supseteq \delta$ by its $k+1$ children

$$c_p \sigma = \sigma \text{ with } \sigma[p] \text{ deleted and } v \text{ inserted at position } l, \quad p \in \{p_0, \dots, p_k\},$$

so that

$$\partial_l c_p \sigma = \partial_p \sigma. \tag{1}$$

A *seed* is *strongly compatible* if all its cells have level 0 and $\partial_i \sigma = \partial_j \omega$ implies $i = j$. Equivalently, the seed is balanced and every cell is ordered by color. Examples are Kuhn's cube and Gaifullin's $\mathbb{CP}_{15}^2$.

2 The face scheme

Let $d = 4$ and $\sigma = \langle a, b, c, d, e \rangle$. The proposal assigns to each cell an integer $l(\sigma) \in \{4, 3, 2\}$ (equivalently a type in $\{0, 1, 2\}$), splits the 2-face

$$F(\sigma) = \{0, 1, l(\sigma)\}$$

(that is, $\langle a, b, e \rangle$, $\langle a, b, d \rangle$ or $\langle a, b, c \rangle$), and inserts the new vertex v at position $l(\sigma)$:

l	split face	children
4	$\langle a, b, e \rangle$	$\langle a, b, c, d, v \rangle$, $\langle a, c, d, e, v \rangle$, $\langle b, c, d, e, v \rangle$
3	$\langle a, b, d \rangle$	$\langle a, b, c, v, e \rangle$, $\langle a, c, d, v, e \rangle$, $\langle b, c, d, v, e \rangle$
2	$\langle a, b, c \rangle$	$\langle a, b, v, d, e \rangle$, $\langle a, c, v, d, e \rangle$, $\langle b, c, v, d, e \rangle$

The type of the children. The proposal does not fix l for the children. There are three level-independent choices, and only one of them works (verified).

rule for the children	result on the reflected Kuhn torus $\mathbb{Z}_4^4$ (6144 cells)
same l as the parent	incompatible after one uniform round (12288 of 18432 cells)
$l \mapsto l + 1$ (i.e. $2 \rightarrow 3 \rightarrow 4 \rightarrow 2$)	incompatible after one uniform round
$l \mapsto l - 1$ (i.e. $4 \rightarrow 3 \rightarrow 2 \rightarrow 4$)	compatible (Theorem 2.2)

So, as in Maubach’s scheme, l decreases by one per level and wraps around:

$$l(\sigma) = 4 - (\text{level}(\sigma) \bmod 3).$$

This is Maubach’s rule $\text{type}(\sigma) = d - (\text{level}(\sigma) \bmod d)$ with the *prefix* $\{0\}$ of the split edge $\{0, l\}$ replaced by the prefix $\{0, 1\}$.

Definition 2.1 (face invariants). A 4-mesh satisfies the *face invariants* if it satisfies (OneLevel) and, for adjacent cells with $\partial_i \sigma = \partial_j \omega$,

$$\text{level}(\sigma) = \text{level}(\omega) \implies i = j \text{ or } \{i, j\} \subseteq F(\sigma), \quad (\text{A})$$

$$\text{level}(\sigma) = \text{level}(\omega) + 1 \implies i = l(\omega) \text{ and } j \in F(\omega). \quad (\text{B})$$

In (B), $i = l(\omega)$ is the position of the weld vertex of the finer cell. A strongly compatible seed satisfies the face invariants.

Theorem 2.2. *On a conforming 4-mesh satisfying the face invariants, the Maubach refinement algorithm (MAUBACHSUBDIVIDE in the article), with the split edge $\sigma[0, \text{type}(\sigma)]$ replaced by the split face $\sigma[F(\sigma)]$, the loop over $i \in \{1, \dots, d\} \setminus \{\text{type}(c)\}$ replaced by $i \notin F(c)$, and the new vertex inserted at $l(\sigma)$, terminates and only splits cells of level at most $\text{level}(\sigma)$. It produces a conforming mesh satisfying the face invariants, and it is consistent, alternating and level-homogeneous.*

This is the case $d = 4$, $k = 2$ of Theorem 3.1, which in turn is a special case of Theorem 4.6. The proof is in Section 4.

Empirical invariants. Under uniform and adaptive refinement from the Kuhn torus (up to level 10), the exceptional pairs in (A) are more restricted than (A) allows. With $l = 4$ only $\{i, j\} = \{0, 1\}$ occurs, and with $l \in \{3, 2\}$ only $\{0, l\}$ and $\{1, l\}$ occur. Every shared facet is an equality of ordered simplices. The weaker statement (A) is the one that is preserved inductively, so it is the one to state.

Why the other rules fail. In the language of Section 4, the two rejected rules fail for different reasons.

- With l fixed, (C1) fails: the weld position l belongs to the children’s split face $\{0, 1, l\}$, so a child would have to split a face containing its own weld vertex.
- With $l \mapsto l + 1$, (C1) holds but (C2) fails. For $l = 3$, the children $\langle b, c, d, v, e \rangle$ and $\langle a, b, c, v, e \rangle$ (deleting a and d) share their facet at positions 2 and 0. That pair is not contained in the next split face $\{0, 1, 4\}$.

3 The prefix family

For $1 \leq k \leq d-1$ and $l \in \{k, \dots, d\}$ put

$$F_l = \{0, 1, \dots, k-1\} \cup \{l\}, \quad l(\sigma) = d - (\text{level}(\sigma) \bmod (d-k+1)),$$

split $\sigma[F_{l(\sigma)}]$, a k -face, and insert the new vertex at $l(\sigma)$. The case $k=1$ is Maubach bisection, and $k=2, d=4$ is Section 2. For $k=d-1$ there are two types, splitting the facets ∂_{d-1} and ∂_d alternately.

Theorem 3.1. *For every $d \geq 2$ and $1 \leq k \leq d-1$, the prefix scheme satisfies the conclusions of Theorem 2.2, with invariants (A) and (B) for $F(\sigma) = F_{l(\sigma)}$.*

Proof. We check conditions (C0)–(C2) of Theorem 4.6. (C0) holds because $l = \max F_l$. For (C1), the next split face is $F_{l'} = \{0, \dots, k-1\} \cup \{l'\}$ with $l' = l-1$, or $l' = d$ if $l = k$. Since $l \geq k$ and $l' \neq l$ (there are $d-k+1 \geq 2$ types), $l \notin F_{l'}$. For (C2), let $p < q$ in F_l and compute $\text{pos}_p(q)$ and $\text{pos}_q(p)$ as in (2).

- If $q < k$, the pair is $\{q-1, p\}$, contained in $\{0, \dots, k-2\} \subseteq F_{l'}$.
- If $q = l$, the pair is $\{l-1, p\}$ with $p \leq k-1$. When $l > k$, $l-1 = l' \in F_{l'}$. When $l = k$, $l-1 = k-1$ lies in the prefix.

This proves the theorem. It was also verified exhaustively for all $d \leq 7$ and all k by checking the two-cell lemma symbolically. $\square$

4 Typed cycles

Definition 4.1. A *typed cycle* in dimension d is a finite sequence $T = (\tau_0, \dots, \tau_{n-1})$ of pairs $\tau = (F_\tau, l_\tau)$, with $F_\tau \subseteq \{0, \dots, d\}$, $|F_\tau| \geq 2$, and $l_\tau \in \{0, \dots, d\}$. A cell σ has type $\tau(\sigma) = \tau_{\text{level}(\sigma) \bmod n}$. It is refined by the ordered stellar subdivision of $\sigma[F_{\tau(\sigma)}]$ with the new vertex at position $l_{\tau(\sigma)}$. We write $F(\sigma) = F_{\tau(\sigma)}$ and $l(\sigma) = l_{\tau(\sigma)}$, and denote by τ^+ the type after τ .

A typed cycle may mix edge types ($|F| = 2$), face types ($|F| = 3$) and so on. Maubach bisection is $((\{0, d\}, d), (\{0, d-1\}, d-1), \dots, (\{0, 1\}, 1))$. The invariants (A) and (B) of Theorem 2.1 make sense verbatim for any typed cycle.

For $p \neq q$, let

$$\text{pos}_p(q) = r + [r \geq l], \quad r = q - [p < q], \quad (2)$$

be the position of $\omega[q]$ in the child $c_p\omega$.

Definition 4.2. A typed cycle is *admissible* if for every type $\tau = (F, l)$:

(C0) not $\min F < l < \max F$;

(C1) $l \notin F_{\tau^+}$;

(C2) for all $p < q$ in F , either $\text{pos}_p(q) = \text{pos}_q(p)$ or $\{\text{pos}_p(q), \text{pos}_q(p)\} \subseteq F_{\tau^+}$.

The conditions have the following meaning. (C0) makes two siblings meet along the same *ordered* facet. (C1) is the alternating property, and it drives the direction lemma. (C2) says that two siblings satisfy (A).

Lemma 4.3 (direction). *Assume (C1) and the invariants, and let $\partial_i\sigma = \partial_j\omega$. If $i \notin F(\sigma)$ then $\text{level}(\omega) \leq \text{level}(\sigma)$, and if $i \in F(\sigma)$ then $\text{level}(\omega) \geq \text{level}(\sigma)$. If the levels are equal and $i \notin F(\sigma)$, then $i = j$ and $\sigma[F(\sigma)] = \omega[F(\omega)]$.*

Proof. If ω is finer, (B) for (ω, σ) gives $i \in F(\sigma)$. If ω is coarser, (B) gives $i = l(\omega)$, and $l(\omega) \notin F(\sigma)$ by (C1), because $\tau(\sigma) = \tau(\omega)^+$. At equal levels with $i \notin F$, (A) forces $i = j$. The two cells then differ only at position $i \notin F$, so they have the same split face. $\square$

Lemma 4.4 (siblings). *Let ω have type $\tau = (F, l)$ and let $p < q$ be in F . Under (C0), the children $c_p\omega$ and $c_q\omega$ share the facet $\partial_{\text{pos}_p(q)}c_p\omega = \partial_{\text{pos}_q(p)}c_q\omega$ as ordered simplices. Under (C2), this pair satisfies (A).*

Proof. Both children contain v and $\omega \setminus \{\omega[p], \omega[q]\}$. The vertices of ω keep their relative order, so the two ordered facets can differ only in where v sits. The vertex v sits at position l in both children, so its predecessors are the first l entries of ω with $\omega[p]$ (respectively $\omega[q]$) removed.

- If $l \geq \max F$, these are $\{\omega[0..l]\} \setminus \{\omega[p]\}$ and $\{\omega[0..l]\} \setminus \{\omega[q]\}$.
- If $l \leq \min F$, both are $\{\omega[0..l-1]\}$.

In either case they agree after removing $\omega[q]$ and $\omega[p]$ respectively. If instead $p < l < q$, the vertex $\omega[l]$ precedes v in one facet and follows it in the other. The indices in the two children are $\text{pos}_p(q)$ and $\text{pos}_q(p)$, and the children have type τ^+ , so (C2) is exactly (A). $\square$

Lemma 4.5 (two cells). *Let σ, ω be adjacent with $\partial_i\sigma = \partial_j\omega$ and $0 \leq \text{level}(\sigma) - \text{level}(\omega) \leq 1$, and let them satisfy the invariants. Let $\tau = (F, l)$ be the type of ω . Split ω , and also σ if it contains $\omega[F]$. Then every adjacent pair formed by a child of ω and σ or a child of σ satisfies the invariants.*

Proof. There are three cases.

- (a) *Equal levels, $j \notin F$.* By Theorem 4.3, $i = j$ and σ has the same split face, so both cells are split. The children $c_p\sigma$ and $c_p\omega$ come from the same positional operation applied to two cells that differ only at position i , so they differ only at position $\text{pos}_p(i)$. This satisfies (A) with equal indices. Children with different p share at most $d-1$ vertices.
- (b) *Equal levels, $j \in F$.* The facet omits $\omega[j] \in \omega[F]$, so σ does not contain the split face and is not split. By (1), $\partial_j\omega = \partial_l c_j\omega$. The pair $(c_j\omega, \sigma)$ has levels $\ell+1, \ell$. The finer index is $l = l(\sigma)$, and the coarser index is $i \in F(\sigma)$ by (A), since $j \in F$ and either $i = j$ or $\{i, j\} \subseteq F$. This is (B).
- (c) *level(σ) = level(ω) + 1.* By (B), $i = l$ and $j \in F$, and only ω is split. Now $\partial_l\sigma = \partial_j\omega = \partial_l c_j\omega$, with equal levels and equal indices: (A).

(OneLevel) holds in every case. $\square$

It remains to see that there are no other new adjacencies. A facet of $c_p\omega$ either omits v , in which case it is $\partial_l c_p\omega = \partial_p\omega$, the outer facet treated in (b) and (c); or it contains v and omits some $\omega[s]$. If $s \in F$, the neighbor across it is the sibling $c_s\omega$ (Theorem 4.4). If $s \notin F$, the facet contains all of $\omega[F]$ except $\omega[p]$, and the neighbor is the child $c_p\omega'$ of the cell ω' across $\partial_s\omega$, which lies in the star of the split face. This is case (a).

Theorem 4.6. *Let T be an admissible typed cycle and K a conforming mesh satisfying (OneLevel), (A) and (B), for example a strongly compatible seed. Then the generalized Maubach algorithm (Algorithm 1) terminates, only splits cells of level at most $\text{level}(\sigma)$, and preserves conformity and the invariants. It is consistent, alternating and level-homogeneous.*

Proof. The proof of the Maubach theorem of the article (§ Maubach's scheme) carries over word for word, with Theorems 4.3 to 4.5 in place of its direction, children and two-cells lemmas. We only list the replacements.

Algorithm 1 CYCLESUBDIVIDE(K, σ) (MAUBACHSUBDIVIDE with F)

```

1:  $\epsilon \leftarrow \sigma[F(\sigma)]; st \leftarrow []; Q \leftarrow (\sigma)$ 
2: while  $Q \neq \emptyset$  do
3:    $c \leftarrow \text{pop}(Q)$ 
4:   if  $c$  not visited then
5:     mark  $c$  visited; append  $c$  to  $st$ 
6:     for  $i \in \{0, \dots, d\} \setminus F(c)$  do ▷ facets of  $c$  containing  $\epsilon$ 
7:        $c_a \leftarrow$  the cell across  $\partial_i c$  (skip if none)
8:       if  $\text{level}(c_a) < \text{level}(c)$  then
9:         CYCLESUBDIVIDE( $K, c_a$ );  $c_a \leftarrow$  the cell across  $\partial_i c$ 
10:      end if
11:      if  $c_a$  not visited then push  $c_a$  onto  $Q$ 
12:    end for
13:  end if
14: end while
15: apply  $(\epsilon, v)_{l(\sigma)}$  to every cell of  $st$ 

```

- The facets explored by the breadth-first search are those that contain the split face, $\partial_i c$ with $i \notin F(c)$.
- The neighbor across such a facet is never finer (Theorem 4.3).
- A coarser neighbor ω meets c along $\partial_j \omega$ with $j \in F(\omega)$, which passes whole to the child $c_j \omega$ (case (c)).
- Alternation is (C1).

□

Necessity (verified, not proved). In dimension 4 we ran an exhaustive symbolic check of the sibling and two-cell configurations allowed by (A) and (B). Admissibility agrees with this check on all 125 one-type and 15625 two-type cycles over $|F| \in \{2, 3, 4\}$, and on 200000 random cycles of length 3–5. So, for the invariants (A) and (B), conditions (C0)–(C2) are not only sufficient but also necessary for these to be preserved. Other invariants could in principle admit other schemes.

5 Geometry

5.1 Faces alone never shrink cells

Proposition 5.1. *If every type of a typed cycle has $|F| \geq 3$, no edge of the mesh is ever removed. In particular, the edges of the seed persist at every level, and the maximal cell diameter does not tend to zero.*

Proof. The stellar subdivision (δ, v) removes exactly the simplices containing δ . An edge does not contain a face δ with at least three vertices. □

For the face scheme on Kuhn’s cube with v at the centroid of the face, the child $\langle a, c, d, e, v \rangle$ of a level-0 cell keeps the diagonal ae . The minimal quality vol/h^4 falls by a factor 3 per level (verified): 0.33, 0.11, 0.037, $\dots$, 0.0014 at level 6, relative to level 0. In the Riemann-surface application, the cells around the curve become needles along seed edges and never get small.

5.2 What to ask of a scheme on $\mathbb{CP}^2$

On Kuhn’s cube, Maubach bisection has finitely many similarity classes. This is *not* the criterion that matters on $\mathbb{CP}^2$: with Fubini–Study midpoints no two cells are similar, and the classes are infinite even for Maubach. What we want is:

- (G1) *Shrinking*: the maximal diameter h_ℓ at level ℓ tends to zero at a uniform rate. A necessary combinatorial condition is a finite *edge lifetime*: there is an L such that no edge of a cell survives in its descendants L levels below.
- (G2) *Shape regularity*: the quality of the cells stays bounded below, or at least degrades slowly.

The flat test on Kuhn’s cube is still the right *necessary* test. At fine levels a cell of a smooth triangulation of $\mathbb{CP}^2$ lies in a region where the Fubini–Study metric is flat up to $O(h^2)$, and geodesic midpoints are affine midpoints up to $O(h^2)$. A scheme whose descendants degenerate in the flat model therefore degenerates on $\mathbb{CP}^2$ as well. Bounded quality on Kuhn’s cube, and not finiteness of the classes, is the property that transfers. We measure the *shrinking exponent* α defined by $h_\ell \approx N_\ell^{-\alpha}$, where N_ℓ is the number of descendants of one cell. The ideal is $\alpha = 1/d$: Maubach has $\alpha = 1/4$ exactly, since each cycle doubles the resolution with 2^4 cells.

5.3 How a hybrid works

A hybrid is an admissible typed cycle that contains both edge types and face types. Refinement works as in Maubach’s algorithm:

- Each cell knows its level, hence its type $\tau = (F, l)$.
- To refine σ , collect the star of the split simplex $\sigma[F]$, recursively refining coarser neighbors first.
- Apply one stellar subdivision with the new vertex at position l .

An edge step splits the star of an edge into 2 children per cell. A face step splits the star of a triangle, which in a 4-manifold is a cycle of cells since the link of a triangle is a circle, into 3 children per cell. The new vertex goes at the midpoint of the edge or at the centroid of the triangle. On $\mathbb{CP}^2$ we use the Fubini–Study barycenter. The motivation for the Riemann-surface application is that face steps split exactly the kind of simplex that is tested against the curve.

The *pointerless* (LPT) encoding of the article carries over. A cell is still a root plus a path of child indices, now with $|F_\tau|$ possible children at a step of type τ . The neighbor-finding rules must be rederived from (1) and Theorem 4.4.

5.4 Search results in dimension 4

All numbers below are from uniform refinement of one Kuhn simplex, with midpoints and centroids. Descendants are deduplicated by similarity, keeping the largest scale. This is exact while the number of classes is below the cap, and a random sample above it. Sampling can only *overestimate* the minimal quality, so the degradation is at least as bad as reported.

- *Cycles of length ≤ 3 mixing $|F| = 2$ and $|F| = 3$* . 2184 are admissible (48 of length 2, 2136 of length 3). Of these, 1392 have finite edge lifetime (4 to 7 levels; Maubach has 3 to 5). None comes close to Maubach: after 12 levels the best has $\alpha \approx 0.11$ and minimal quality below 0.003. This is expected, since a cycle of length ≤ 3 cannot contain the 4 edge types that Maubach needs.

- *Maubach with one face step.* Inserting one arbitrary face type into Maubach’s 4-cycle is never admissible. Replacing one edge type is admissible in exactly 3 ways, all of which replace $(\{0, 2\}, 2)$ by $(\{0, 1, 2\}, l)$ with $l \in \{2, 3, 4\}$. The best is

$$T^* = ((\{0, 4\}, 4), (\{0, 3\}, 3), (\{0, 1, 2\}, 2), (\{0, 1\}, 1)),$$

a variable-prefix scheme ($k = 1, 1, 2, 1$). Per cycle of 24 cells:

cycle	1	2	3	4	5	6
h , Maubach (16× cells/cycle)	0.500	0.250	0.125	0.062	0.031	0.016
$q_{\min}$, Maubach	1	1	1	1	1	1
h , T^* (24× cells/cycle)	0.500	0.276	0.176	0.108	0.068	0.039
$q_{\min}$, T^*	0.667	0.298	0.075	0.022	0.006	0.002

Cells do shrink (finite edge lifetime, $\alpha \approx 0.18$), but the minimal quality falls by a factor of about 3–4 per cycle, so (G2) fails.

5.5 Conclusions and open questions

1. The face scheme is a correct stellar subdivision scheme, and so is the whole admissible typed-cycle family. The combinatorics of compatibility do not care whether the split simplex is an edge or a face.
2. Geometrically, pure face schemes are ruled out (Theorem 5.1), and the hybrids we tested degenerate. Edge bisection is therefore needed for the *geometry*, not for compatibility.
3. Open questions:
 - longer hybrid cycles (length ≥ 5 , several face steps per Maubach cycle);
 - placing the face vertex somewhere other than the centroid, which cannot help pure face schemes (Theorem 5.1) but might help hybrids;
 - invariants other than (A) and (B).
4. For the Riemann-surface application, the practical recommendation is to keep Maubach bisection for refinement and use the 2-faces in the refinement *criterion*, which is what the code already does.

Scripts

scripts/ (Python 3, NumPy):

- `facesplit.py`: mesh simulator on the reflected Kuhn torus, with uniform and adaptive refinement.
- `inv.py`: census of the $(\Delta\text{level}, \text{type}, i, j)$ pairs.
- `symbolic.py`: two-cell check for the prefix family, $d \leq 7$.
- `hybrid.py`: search for admissible edge+face cycles.
- `charac_rule.py`, `charac.py`: conditions (C0)–(C2) and their agreement with the exhaustive check.
- `lifetime.py`: edge lifetime.
- `geo.py`, `geo2.py`, `screen.py`, `insert.py`, `traj.py`: geometry on Kuhn’s simplex.